



Department of
Software Engineering
BAHRIA UNIVERSITY
Discovering Knowledge

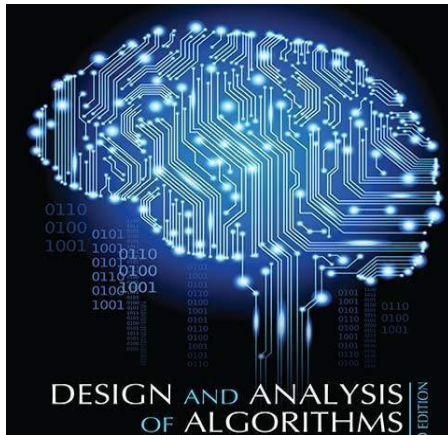
CSC - 321

DESIGN AND ANALYSIS OF ALGORITHMS

COURSE INSTRUCTOR

Dr. Qamaruddin Memon

Email: Qamaruddin.bukc@bahria.edu.pk



LECTURE 4

Recurrence

OUTLINE

- Recurrence
- Fibonacci Sequence
- Fibonacci Problem and its Sequence
- Construction of Mathematical Model
- Explicit Formula Computing Fibonacci Numbers Recursive Algorithms
- Generalizations of Rabbits Problem and Constructing its Mathematical Models Applications of Fibonacci
- Sequences

RECURSION

It is a method of solving a problem where the solution depends on solutions to smaller instances of the same problem.

$$f(x) = 2f(x - 1) + x^2$$

$$f(0) = 0 \text{ (Base)}$$

$$f(1) = 1,$$

$$f(2) = 6,$$

$$f(3) = 21$$

RECURSIVE METHOD

A method that is defined in terms of itself is called **recursive**.

Recursive method *f* ought to be expressible in only a few lines, just like a non-recursive method .

Recursive calls will keep on being made until a base case is reached

Base Case: The value for which the function is directly known without resorting to recursion

RECURSIVE METHOD

$$f(x) = 2f(x - 1) + x^2$$

```
public static int f( int x )  
{  
    if( x == 0 )  
        return 0;  
    else  
        return 2 * f( x - 1 ) + x * x;  
}
```

What will happen for ***f(-1)***?

ANALYZING RECURSIVE METHODS

Factorial: $5! \rightarrow \text{fact}(5) \rightarrow 5.4.3.2.1$

If the recursion is really just a thinly veiled for loop, the analysis is usually trivial. For instance, the following method is really just a simple loop and is $O(N)$:

```
public static long factorial( int n )  
{  
    if( n <= 1 )  
        return 1;  
    else  
        return n * factorial( n - 1 );  
}
```

ANALYZING RECURSIVE METHODS

Fibonacci Series: 1, 1, 2, 3, 5, 8, ...

```
public static long fib( int n )  
{  
    if( n <= 1 )  
        return 1;  
    else  
        return fib( n - 1 ) + fib( n - 2 );  
}
```

RECURRENCE RELATION

A recurrence relation is an equation which is defined in terms of itself

Many algorithms are recursive in nature. When we analyze them, we get a recurrence relation for time complexity.

When it is difficult to convert the recursion into a simple loop structure, the analysis will involve a **recurrence relation** that needs to be solved.

$$T(n) = 2T(n/2) + cn$$

No general procedure for solving recurrence relations is known which is why it is an art

- Back substitution
- Recurrence Tree etc

ANALYZING RECURSIVE METHODS

```
public static long fib( int n )  
{  
    if( n <= 1 )  
        return 1;  
    else  
        return fib( n - 1 ) + fib( n - 2 );  
}
```

$$T(N) = T(N - 1) + T(N - 2) + 2$$

ANALYZING RECURSIVE METHODS

By induction it can be proved that the running time of this program grows exponentially

By keeping a simple array and using a for loop, the running time can be reduced substantially

This program is slow because there is a huge amount of redundant work being performed [$f(n-1)$ and $f(n-2)$]

It is generally a bad idea to use recursion to evaluate simple mathematical functions, such as the Fibonacci numbers

The main problem with recursion is the hidden bookkeeping costs

RULES OF RECURSION

Four fundamental rules of recursion are:

1. **Base cases:** You must always have some base cases, which can be solved without recursion.
2. **Making progress:** For the cases that are to be solved recursively, the recursive call must always be to a case that makes progress toward a base case.
3. **Design rule:** Assume that all the recursive calls work.
4. **Compound interest rule:** Never duplicate work by solving the same instance of a problem in separate recursive calls

FIBONACCI SEQUENCES (NATURAL MODELS)

FIBONACCI SEQUENCE

By studying Fibonacci numbers and constructing Fibonacci sequence we can imagine how mathematics is connected to apparently unrelated things in this universe.

Even though these numbers were introduced in 1202 in Fibonacci's book *Liber abaci*, but these numbers and sequence are still fascinating and mysterious to people of today.

Fibonacci, who was born Leonardo da Pisa gave a problem in his book whose solution was the Fibonacci sequence as we will discuss it today.

FIBONACCI'S PROBLEM

Statement:

Start with a pair of rabbits, one male and one female, born on January 1.

Assume that all months are of equal length and that rabbits begin to produce two months after their own birth.

After reaching age of two months, each pair produces another mixed pair, one male and one female, and then another mixed pair each month, and no rabbit dies.

How many pairs of rabbits will there be after one year?

Answer: *The Fibonacci Sequence!*

0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, . . .

CONSTRUCTION OF MATHEMATICAL MODEL

$$F_0 = 0$$

$$F_1 = 1$$

end of month 1

end of month 2

$$F_2 = 1$$

end of month 3

$$F_3 = 2$$

$$F_4 = 3$$

end of month 4

$$F_5 = 5$$

end of month 5

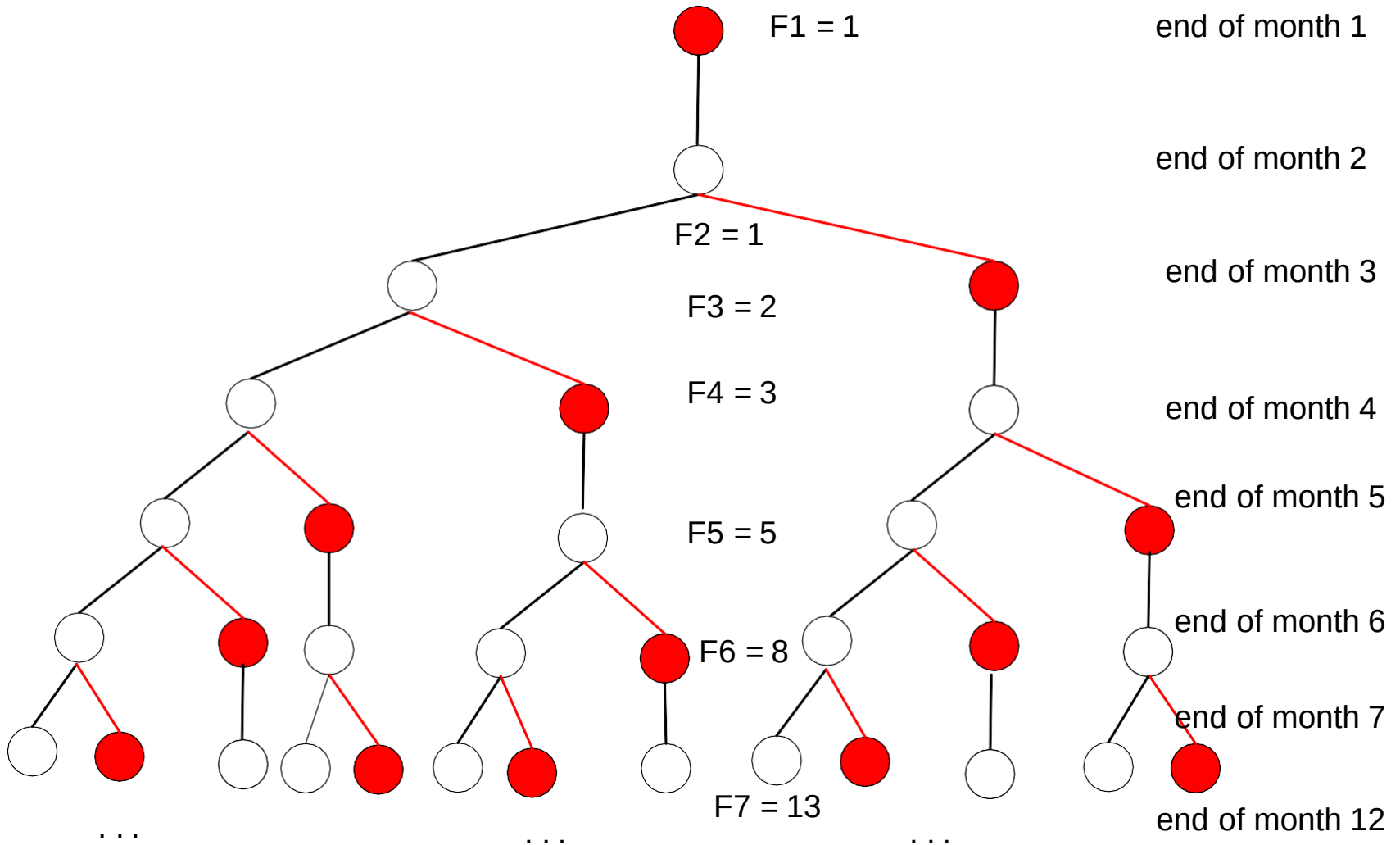
$$F_6 = 8$$

end of month 6

$$F_7 = 13$$

end of month 7

end of month 12



CONSTRUCTION OF MATHEMATICAL MODEL

Total pairs at level k = Total pairs at level $k-1$ + Total born at level k (1)
pairs

Since

Total pairs born at level k = Total pairs at level $k-2$ (2)

Hence by equation (1) and (2)

Total pairs at level k = Total pairs at level $k-1$ + Total at level $k-2$
pairs

Now let us denote
 F_k = total pairs at level k

Now our recursive mathematical model will become

$$F_k = F_{k-1} + F_{k-2}$$

COMPUTING VALUES USING MATHEMATICAL MODEL

Since

$$F_k = F_{k-1} + F_{k-2}$$

$$F_0 = 0, F_1 = 1$$

$$F_2 = F_1 + F_0 = 1 + 0 = 1$$

$$F_3 = F_2 + F_1 = 1 + 1 = 2$$

$$F_4 = F_3 + F_2 = 2 + 1 = 3 \quad F_5 =$$

$$F_4 + F_3 = 3 + 2 = 5 \quad F_6 = F_5 +$$

$$F_4 = 5 + 3 = 8 \quad F_7 = F_6 + F_5 =$$

$$8 + 5 = 13 \quad F_8 = F_7 + F_6 = 13$$

$$+ 8 = 21$$

$$F_9 = F_8 + F_7 = 21 + 13 = 34$$

$$F_{10} = F_9 + F_8 = 34 + 21 = 55 \quad F_{11}$$

$$= F_{10} + F_9 = 55 + 34 = 89$$

$$F_{12} = F_{11} + F_{10} = 89 + 55 = 144 \dots$$

APPLICATIONS OF FIBONACCI SEQUENCES

- Are used in trend analysis
- By some pseudorandom number generators
- The number of petals is a Fibonacci number.
- Many plants show the Fibonacci numbers in the arrangements of the leaves around the stems.
- Seen in arrangement of seeds on flower heads
- Consecutive Fibonacci numbers give worst case behavior when used as inputs in Euclid's algorithm.
- As n approaches infinity, the ratio $F(n+1)/F(n)$ approaches the golden ratio:

$$\phi = 1.6180339887498948482...$$

